

 HOSPITALITY & TOURISM TECH

Digital Transformation

Accelerating Innovation, Smart Operations, and Personalised Experiences in Singapore's Hotel Industry

16.9M

Visitors (2025)

S\$32.8B

Tourism Receipts

ITM 2025

STB Roadmap

SECTION 01

Market Drivers



Why Singapore's tourism and hospitality sector must transform to maintain global leadership.

| Global Competition & Visitor Growth

Regional Competition

Singapore competes aggressively with regional powerhouses like Bangkok, Bali, Kuala Lumpur, Tokyo, and Hong Kong.

Strategic Impact: Hotels must offer differentiated experiences and superior service standards to attract high-yield international travellers.

Escalating Visitor Volume

Welcoming 16.9 million international visitors requires high-capacity operational workflows.

Strategic Impact: Hospitality businesses need automated, scalable infrastructure to manage high guest volumes smoothly without service degradation.

Productivity & Travel Trends

- ✓ **Labour & Productivity Pressures:** Managing workforce constraints while delivering 5-star service drives adoption of AI scheduling and service robotics.
- ✓ **Changing Traveller Expectations:** Guests actively seek wellness, immersive cultural stories, eco-sustainability, and hyper-personalized hospitality.
- ✓ **Digital Platform Competition:** Online travel platforms intensify price transparency, requiring hotels to strengthen direct digital engagement.



SECTION 02

Technology Drivers



The technological enablers making seamless and smart hospitality operations possible.

Key Enabling Technologies



AI & GenAI

AI concierge chatbots provide instant 24/7 guest service, while Generative AI builds custom travel itineraries based on guest preferences.



IoT & Smart Rooms

In-room automated sensors enable guests to control lighting, temperature, and entertainment while optimizing hotel energy consumption.



Contactless Tech

Facial recognition and STB's E-Visitor Authentication (EVA) system streamline check-in, reducing queue times to seconds.

| STB Tech Adoption in Action

15m

Down from 2 Hours

Conrad Centennial Singapore Case Study

By implementing an AI-powered staff scheduling system, the hotel drastically reduced roster planning time from 120 minutes down to just 15 minutes, liberating management for guest interactions.

2025 Benchmark: Over 28 major Singapore tourism enterprises adopted advanced AI & automation solutions with STB support in 2025.

SECTION 03

Customer Drivers



How evolving guest expectations reframe digital touchpoints across the entire travel cycle.

Expectations vs Digital Response

Traveller Expectation	Core Need	Digital Solution Response
Convenience & Speed	"I want zero waiting time at reception."	EVA Facial Recognition Mobile Check-in
Hyper-Personalisation	"Recommend activities tailored to me."	GenAI Itineraries Predictive CRM
24/7 On-Demand Help	"I need instant support anytime."	AI Voice & Chatbots Smart Concierge
Sustainability Focus	"I prefer eco-responsible stays."	IoT HVAC Sensors Energy Analytics

The Connected Guest Journey

1

Pre-Arrival

AI analyzes historical preferences & sends tailored room recommendations.

2

Arrival

Digital self check-in via mobile key and facial recognition bypasses queues.

3

In-Stay

App controls IoT room settings; AI concierge curates local attraction suggestions.

4

Post-Stay

CRM records stay data to dispatch personalized rewards for future stays.

STB & OpenAI Strategic Vision

In mid-2025, the Singapore Tourism Board partnered with OpenAI under an MoU to build an AI-native tourism ecosystem.

🔧 From Generic to Intelligent

Moving beyond static lists ("Top 20 attractions") toward context-aware trip planning tailored to family composition, weather, and venue proximity.



 INTERACTIVE SESSION

Questions & Discussion

Building the Future of Tech-Enabled Hospitality in Singapore

Singapore Tourism Board (STB) Hotel Transformation Roadmap

Image Sources



<https://research.ehl.edu/hubfs/EHL-RESEARCH/Blog%20Header/invisible-technologies.jpg>

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